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Project #34: Initial Clostridioides difficile Testing

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AIM

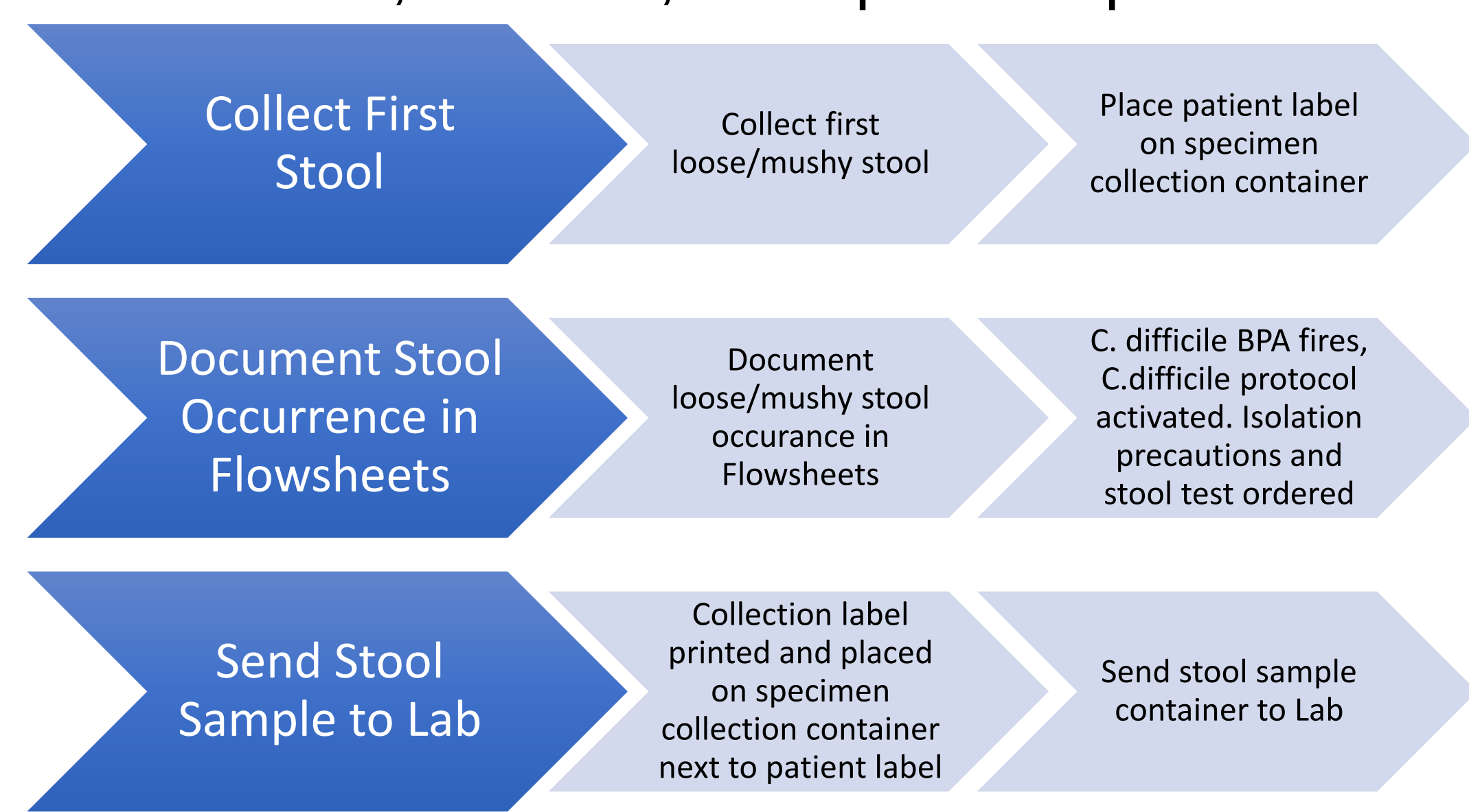
- Problem Statement: First Floor GPU (General Practice Unit) at Henry Ford West Bloomfield Hospital experienced an increased number of hospital acquired *Clostridioides difficile* infections in Q1 and Q2 of 2023. Many of those cases occurred because there was a delay in testing due to length of time between stooling and testing within the first three days of hospital admission, which is the community onset window
- Improvement Statement: Design new stool collection protocol to decrease missed opportunities to test for *Clostridium difficile* within community onset window

PLAN: CURRENT STATE

- Currently, testing for *Clostridioides difficile* (*C. difficile*) is collected after the first documented loose/mushy stool within the first three days after inpatient admission
- When the first loose/mushy stool is documented *Clostridium difficile* Best Practice Alert (BPA) fires and protocol is activated
- There have been cases when the patient will not have another stool occurrence for days after the protocol was activated and stool sample cannot be tested. If testing occurred after the community onset window, which is after the third day of admission, it is considered hospital acquired *C. difficile* infection

DO: CORRECTIVE ACTIONS/INTERVENTIONS

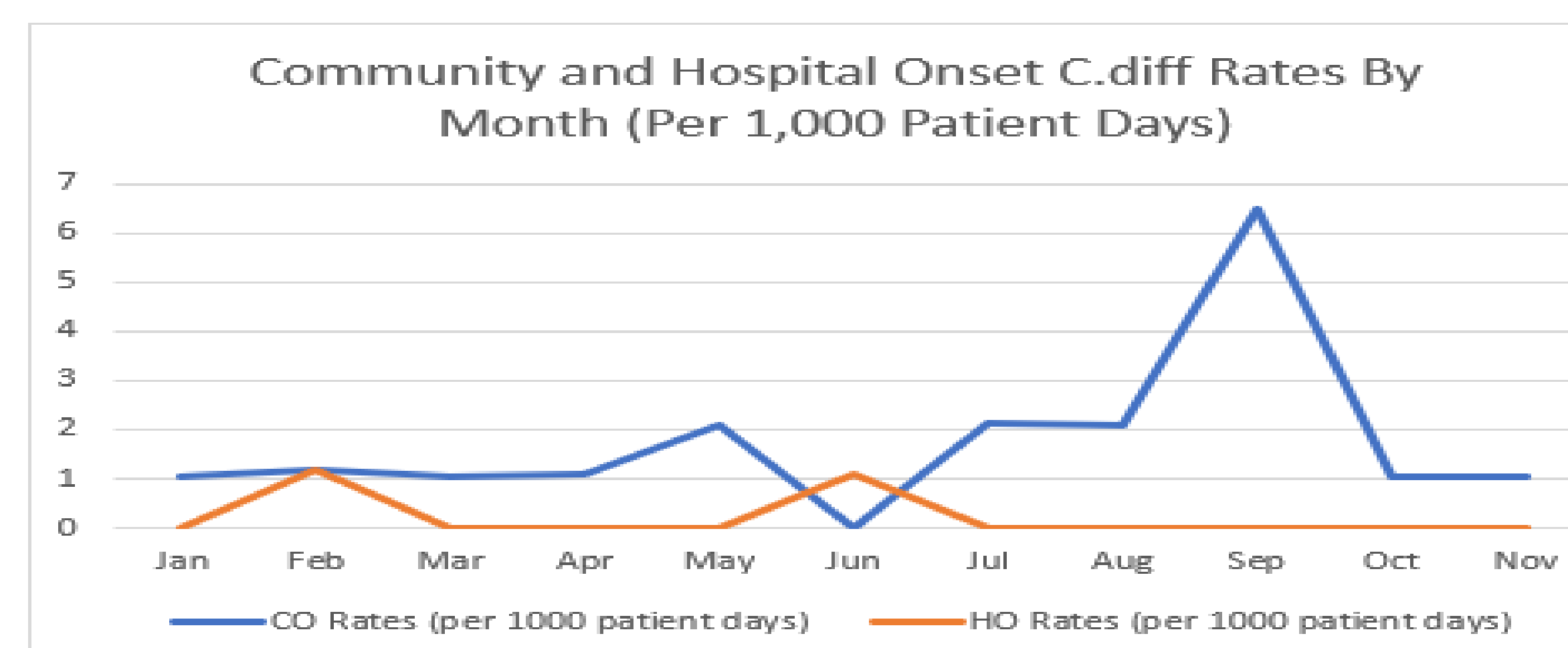
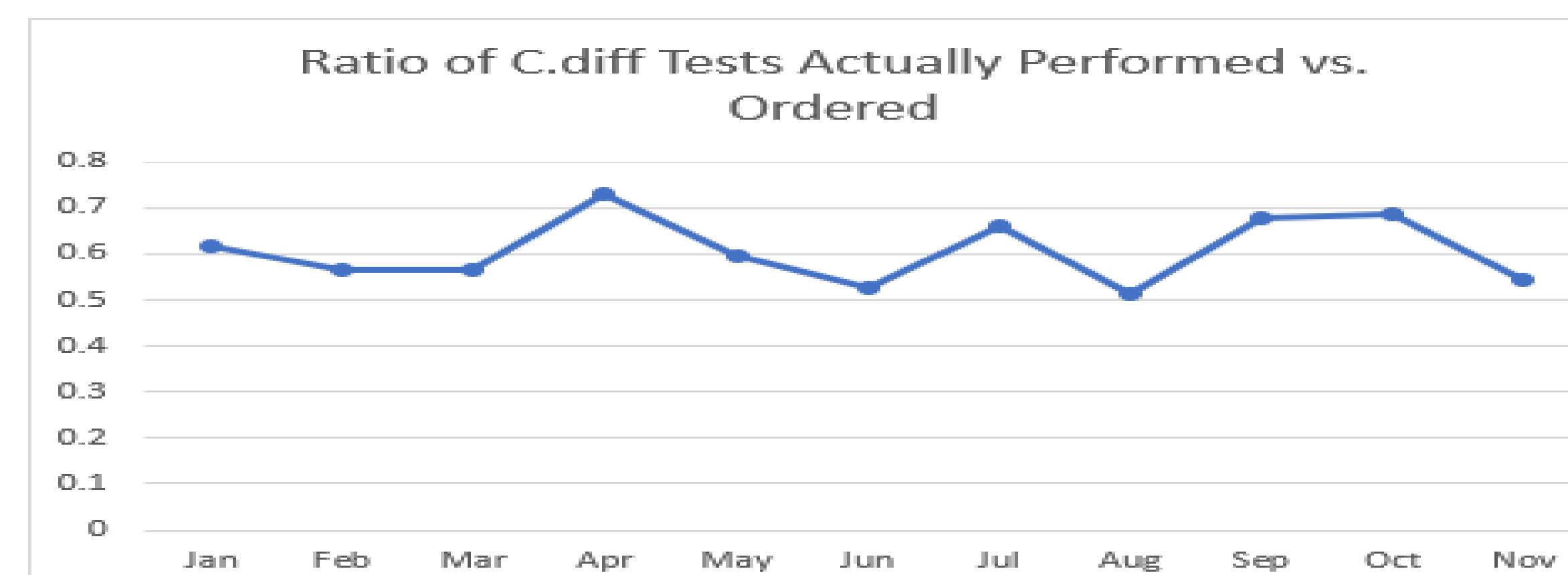
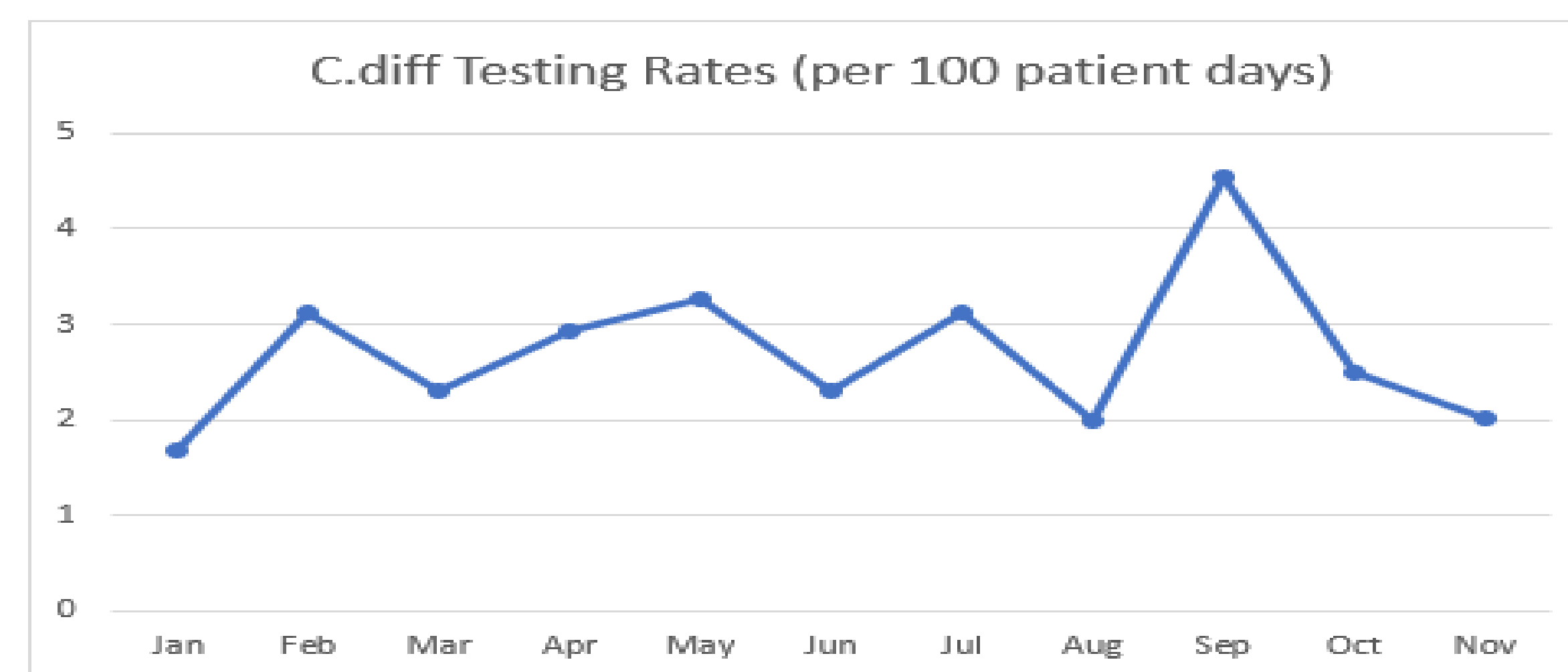
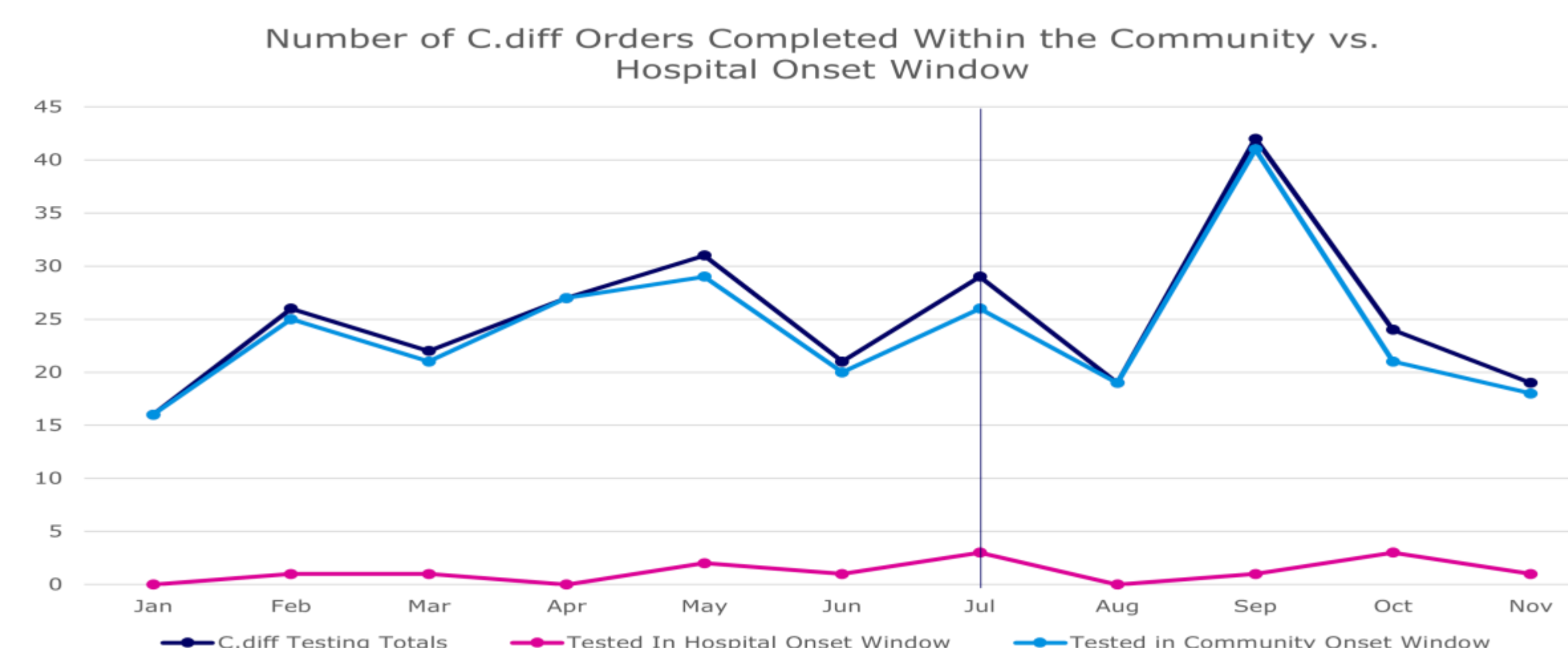
- Development of new *C. difficile* stool collection protocol and was implemented during Q3 and Q4 to improve capturing community onset infections
- New process consists of collecting the first loose/mushy stool, then document stool occurrence in Flowsheets, activation of *C. difficile* protocol, test ordered and sent to Lab
- Education developed by nurse educator that included Infection prevention specialists and unit leadership
- Education provided to nurses and nursing assistants through practice alerts, huddles, and tipsheets posted on unit



CHECK (EVALUATION OF CHANGES)

- During the third and fourth quarter of stool collection and testing, audits were performed, which revealed an increase in the number of tests conducted within the community onset timeframe
- Increased number of first stool sample collection and testing was completed within the first three days after inpatient admission(community onset), and decreased number of collection and testing after the third day (hospital onset).
- The number of community onset rates increased, and hospital onset rates decreased.
- Missed opportunities to test avoided

MEASURES



ACT: SUSTAIN AND SPREAD

- Daily audits were completed, and findings shared during safety huddle
- New collection process posted in the department at both nursing stations as a reference for both nurses and nursing assistants.
- Monthly data reviews were conducted, and front-line staff was interviewed about the effectiveness of the new stool collection process.
- Overall Hospital acquired *C. difficile* rates have shown significant reduction due to early stool collection and testing.
- Towards the end of the fourth quarter and after evaluating the data and seeing reduction in hospital acquired *C. difficile* on First Floor GPU, the process was adopted throughout HFWB hospital.

ADDITIONAL PHOTOS OR VISUALS

New *C.difficile* Stool Sample Collection and Test Ordering Process

- Currently, when the patient has loose/mushy stool, we document the occurrence, and then put test order in for collection.
- There have been several hospital onset C.diff cases where the patient may not have loose/mushy/watery stools for days following the initial loos stool, leading to missed opportunity.
 - In several other cases, the patient only had loose/mushy/watery stools documented after the 3rd hospital day (excludes ED or CDU) even though the patient had loose/mushy/watery stools several times within the prior 3 hospital days.
- For those reasons, we started a new process for ordering the C-diff testing and collecting C.diff stool samples.
- When the patient has the first loose/mushy/watery stool, the stool must be collected then if possible. This includes instances such as loose/mushy/watery bowel movements that may occur during patient clean up.
- DO NOT wait to collect the stool sample after documenting the occurrence in Epic, the test to be ordered, or for the patient to have another stool.**
- After collecting the stool sample, place patient label on the collection container, then document the loose/mushy stool on the Flowsheets and using the C-diff BPA and protocol order the test in EPIC. After printing the collection manager label place, it on the collection container where the patient label is. **DO NOT print the label before the stool sample has been collected.**
- If the patient has loose/mushy/watery stool on inpatient hospital day 4 (excludes ED and CDU), you must contact the provider or Infection Prevention to see if it is appropriate for a C-diff to be ordered.
- The count for the first hospital day begins when the patient physically gets to the inpatient unit (excludes ED and CDU). It doesn't matter how much time the patient has been on the unit on that hospital day. If the patient is under observation status, but physically on an inpatient unit, the patient is still included for C-diff testing if the criteria has been met. Example: If a patient is admitted to 1GPU at 23:58 on 7/13, 7/13 is still considered hospital day 1.

Figure 1: New Process Education Tipsheet Posted on the Unit

KEYS TO SUCCESS/LESSONS LEARNED

- Multidisciplinary team worked together identifying the gaps for missed testing opportunities and developed a plan to improve testing within the community onset window
- Dedicated staff that understood and believed in the importance of testing the first stool occurrence within the community window was a key to success
- Hospital acquired *C. difficile* rates decreased likely due to early stool collection and testing
- Tracking and sharing data regularly will ensure that desired results are achieved